

Miray Hirabayashi

(515)-230-2339 | mirayhirabayashi@gmail.com | linkedin.com/in/miray-hirabayashi | github.com/MirayHirabayashi

EDUCATION

Iowa State University

Ames, Iowa

Bachelor of Science, Major in Software Engineering, Minor in Cybersecurity

Expected Graduation: May 2027

- **Relevant Coursework:** Object-Oriented Programming, Data Structures, Algorithms, Digital Logic, Cybersecurity Concepts, Embedded Systems, UI Design, Software Development Practices, Database Management Systems, Software Testing

WORK EXPERIENCE

Iowa Department of Transportation

Ames, Iowa

Administrative Intern

May 2025 - Present

- Developed a **Python** CLI application to automate validation and reconciliation between Workday and Microsoft Access datasets, processing **11,000+** records and eliminating manual verification.

Iowa State University Language Studies Resource Center

Ames, Iowa

Student Assistant

Dec 2024 - Present

- Provided technical support and troubleshooting for hardware and software used by students and faculty.

PROJECTS

NextClose – Stock Movement Predictor | Python, FastAPI, scikit-learn, Docker, Google Cloud Platform (Cloud Run), Next.js, TypeScript, PostgreSQL, React, Tailwind CSS

- Built a full-stack ML web application that predicts next-day stock price direction using a random forest classifier trained on **35+** years of Yahoo Finance data, exposing predictions through a **FastAPI** REST API and a server-rendered **Next.js** frontend.
- Containerized the ML backend with **Docker** and deployed it to **Google Cloud Run**, enabling serverless auto-scaling through environment-based configuration and cloud-native deployment.
- Engineered a **PostgreSQL** caching layer on **Supabase** with pooled connections and idempotent upserts keyed to the latest market close, reducing repeat prediction latency from **~50** seconds of model retraining to near-instant cache retrieval.

GlobeBase – Interactive 3D Travel Tracker | Next.js, React, TypeScript, Prisma, PostgreSQL, Three.js, Tailwind CSS

- Built a full-stack 3D globe travel tracker rendering **250+** geographic entities using **WebGL (Three.js)** with a hybrid server-side rendered (SSR)-safe **Next.js** App Router architecture.
- Designed optimistic UI updates with SWR cache synchronization across 4 typed REST API routes validated via **Zod**, enabling instant cross-component state consistency.
- Engineered **Prisma**-based **PostgreSQL** upsert logic to eliminate manual seeding for **195** countries, and integrated **Supabase Storage** to persist user travel state across sessions and devices.

WorkSync – Android App | Java, MySQL, Android Studio, Gradle

- Developed an employee and project management application for small businesses to track staff, assign tasks, and manage workflows.
- Built Android UI and application logic in **Java**, integrating REST APIs, WebSockets, and full CRUD functionality.
- Collaborated in an Agile team using **Git/GitLab** with feature branches and merge requests, implemented CI/CD pipelines, and developed unit tests to improve code quality and deployment reliability.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C, PHP, SQL, Swift, HTML, CSS
- **Frameworks & Libraries:** React, Angular, Next.js, Node.js, Express, Flask, Prisma, Tailwind CSS, SwiftUI, scikit-learn
- **Cloud & DevOps:** Google Cloud Platform (GCP), Docker, Supabase, Vercel
- **Developer Tools:** Git (GitHub, GitLab), Postman, Figma, Xcode, Android Studio
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Operating Systems & Environments:** Linux (Ubuntu, Debian, Kali), macOS, Windows